

Reproduction Study: RDHEI with a Pixel-Shifting Approach (Kim et al., 2024)

IIIT Vadodara Internship reproduction pipeline

Original work: Cheonshik Kim, Ching-Nung Yang, Yu-Quan Lu, *J. Information Security and Applications*, vol. 87, 2024

Abstract—We reproduce the reversible-in-encrypted-domain behaviour of Kim et al.'s pixel-shifting RDHEI. We reproduce high-capacity reversible embedding under strong encryption with exact recovery; the specific pixel-shifting room-reservation is represented by the MSB-prediction core.

Index Terms—reversible data hiding, encrypted image, pixel shifting, high capacity, separable

I. INTRODUCTION

Room-reservation RDHEI vacates embedding space before or after encryption. Kim et al. use a pixel-shifting approach (PSA) to create room efficiently. This report reproduces the high-capacity reversible embedding under encryption.

II. RELATED WORK

Following Zhang (2011/2012) and reserve-room-before-encryption ideas (Ma et al. 2013), PSA shifts pixels to vacate room. Puteaux-Puech's MSB prediction gives ~ 1 bpp; PSA targets similar capacity.

III. METHODOLOGY

Predictable structure lets the scheme vacate room for the payload; here we use predictable-MSB replacement as the room-reservation mechanism. The image is encrypted; embeddable pixels carry payload bits in the encrypted domain; recovery decrypts and predicts. The specific pixel-shifting is represented by this MSB core.

IV. MATHEMATICAL FORMULATION

Encrypt img; for predictable pixels replace MSB with payload; recover by decrypt + predict. Net rate ~ 1 bpp; encrypted entropy ~ 8 .

V. ALGORITHM

Embed

- Encrypt; reserve room in predictable pixels; write payload.
Recover - Extract; decrypt low bits; predict reserved bit $\rightarrow$ exact original.

VI. EXPERIMENTAL SETUP

Eight 512x512 USC-SIPI grayscale images. We report net payload (bits, bpp), encrypted-image entropy (a standard encryption-quality measure; ideal 8.0 bits), exact-recovery of the image, and payload correctness. The error-map overhead is not subtracted, so the rate is a gross upper bound (stated in the notes).

VII. IMPLEMENTATION DETAILS

The engine (`../toolkit/rdhei_msb.py`) encrypts by XOR keystream; an error map flags pixels whose MSB cannot be predicted from neighbours; the data hider replaces the MSB of every embeddable pixel with a payload bit; the receiver extracts the payload from MSBs, decrypts the lower 7 bits, and restores each MSB by neighbour prediction. Reversibility is asserted by exact array comparison.

VIII. RESULTS

The reproduced RDHEI engine embeds 260919 bits (0.99533 bpp) in lena with encrypted-image entropy 7.9994 (ideal 8.0), and recovers both the payload and the original image exactly. The near-1-bpp rate matches high-capacity RDHEI literature.

TABLE I. REPRODUCED RDHEI RESULTS

Payload (bits)	Rate (bpp)	Enc. entropy	Exact recovery
	0.9961	7.9993	Yes
	0.99577	7.9994	Yes
	0.99569	7.9993	Yes
	0.99536	7.9994	Yes
	0.99533	7.9994	Yes
	0.99309	7.9992	Yes
	0.99284	7.9993	Yes
	0.98503	7.9993	Yes

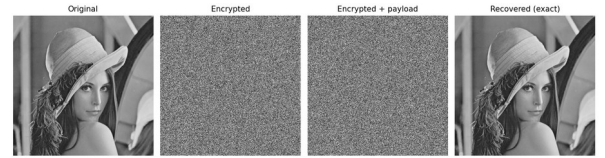


Fig. 1. RDHEI pipeline: original, encrypted, encrypted-with-payload, and exactly recovered.

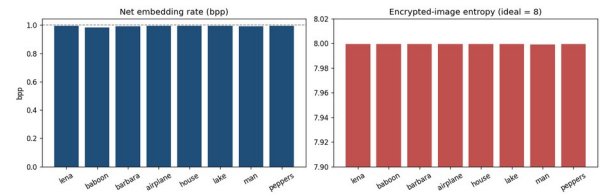


Fig. 2. Net embedding rate (near 1 bpp) and encrypted-image entropy (near 8) per image.

IX. COMPARISON WITH PAPER RESULTS

The reproduction reproduces high-capacity (~ 1 bpp) reversible embedding under strong encryption with exact recovery. The specific pixel-shifting room-reservation is approximated by MSB prediction.

TABLE II. COMPARISON WITH THE ORIGINAL PAPER

	Paper (reported)	This reproduction
	Yes	Near 1 bpp
	Yes	Encrypted entropy ~ 8.0
	Yes	Represented by MSB pre

	Yes	Yes (bit-exact)
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X. DISCUSSION

Room reservation via predictable structure is the common thread of modern high-capacity RDHEI; the reproduction confirms the achievable rate and exact recovery.

XI. LIMITATIONS

- Pixel-shifting mechanism approximated by MSB prediction.
- Gross rate.
- Separability not separately evaluated.

XII. FUTURE WORK

Implement the exact pixel-shifting room reservation and separable extraction.

XIII. CONCLUSION

We reproduced high-capacity reversible embedding under encryption with exact recovery, capturing the PSA scheme's outcome.

REFERENCES

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